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RESEARCH ARTICLE

Development and Validation of an AI-Powered OCR Educational Tool for Early Handwriting Instruction for Korean Early Elementary Students

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ABSTRACT This work developed and classroom-deployed the AI Hangul Writing Teacher, an artificial-intelligence-based optical character recognition tool designed to deliver scalable, individualized feedback in early handwriting instruction. Paper worksheets were photographed by teachers and processed on the server; students in the feedback condition received prompt post-submission reports showing overall accuracy, whereas the control condition operated in a measurement-only mode in which accuracy was computed but not displayed during practice. In a 10-day, class-level randomized classroom study with 91 first-grade students, handwriting recognition accuracy in the feedback group increased from 65% to 95%, compared with 66% to 73% in the control group. Post-test measures of learning flow and academic motivation also favored the feedback group ($p < .001$). These results are consistent with the value of prompt, system-generated feedback for strengthening early handwriting performance and engagement while preserving a paper-based workflow. The study further indicates that well-designed recognition systems can provide effective and scalable support for foundational literacy instruction. Limitations include the single-school setting, the 10-day duration, and provision of overall accuracy only (no per-character highlights). Future work will broaden the evaluation across schools and grade levels, examine more granular formative feedback and long-term retention, and optimize the capture pipeline and teacher-facing tools.

INDEX TERMS Artificial intelligence, educational technology, elementary education, handwriting recognition, optical character recognition.

I. INTRODUCTION

Handwriting remains a foundational skill in primary education, yet providing immediate, individualized feedback to students in large classrooms presents a significant pedagogical challenge [1], [2], [3]. While recent advancements in deep learning, particularly in Optical Character Recognition (OCR), offer powerful technological solutions, their application has predominantly focused on character classification and digitization [4], [5], [6]. A critical research gap exists in the empirical validation of these technologies as real-time instructional tools within authentic educational

environments, especially for complex, multi-stroke scripts such as Hangul, whose syllable blocks assemble jamo and follow normative stroke orders and spatial layout rules [7], [8]. This paper addresses this gap by developing, deploying, and evaluating a novel educational application that leverages a deep learning-based OCR engine to provide prompt, score-based feedback (overall accuracy) after submission within the class session, without per-character highlights, for early Hangul handwriting instruction.

The integration of artificial intelligence (AI) into education, often in the form of Intelligent Tutoring Systems (ITS), has long aimed to personalize learning [9], [10]. Building on this, our work moves beyond traditional AI-in-education frameworks by integrating a custom-trained Convolutional

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Neural Network (CNN) model directly into a user-facing application designed for young learners and optimized for child handwriting captured under classroom conditions (mobile photos with document rectification and layout normalization). Unlike conventional OCR systems focused on accuracy for document processing, our engine is specifically optimized for analyzing the formative stages of character writing, identifying common mistakes made by novices (e.g., vowel slant, batchin omissions, spacing) [11], [12]. This allows the application to serve not just as a recognition tool, but as an application that returns score-based, instruction-aligned feedback for early handwriting practice.

To validate the practical efficacy of our approach, we conducted a 10-day, class-randomized empirical study with first-grade students ($N = 91$) at a public elementary school in Buk-gu, Daegu, South Korea. Prior to data collection, the study was approved by the school principal, and written informed consent for participation and the processing of students' personal data was obtained from parents/guardians; all data were de-identified before analysis. The experiment was designed to quantitatively measure the impact of the application on key educational metrics—handwriting accuracy (0–100), ARCS-based academic interest, and EduFlow-2—and to analyze the perceptions of both students and teachers to gain qualitative insights into the tool's usability, engagement, and potential for integration into standard curricula.

The primary contributions of this paper are therefore three-fold:

- **Development of a Novel Educational Tool:** We designed and implemented a complete application that integrates a specialized, deep learning-based OCR engine to analyze handwritten Hangul characters and provide prompt, score-based feedback (overall accuracy; no per-character highlights) for educational purposes.
- **Rigorous Empirical Validation:** We conducted a controlled, class-randomized study in a real classroom setting to quantitatively verify the effectiveness of the application in improving handwriting accuracy (0–100), ARCS-based interest, and EduFlow-2, comparing Immediate-Feedback with Delayed-Feedback conditions.
- **Practical Insights for AI in Education:** We provide a comprehensive analysis of user perceptions from both students and teachers, offering practical guidance for the successful deployment and adoption of AI-based pedagogical tools in early education.

The remainder of this paper is organized as follows. Section II provides a review of related literature on OCR technology and AI in education. Section III details our methodology, first by describing the design and implementation of our educational system, and second by outlining the experimental design and procedure. Section IV presents the quantitative and qualitative results of our study. Finally, Section V discusses the implications of our findings and offers concluding remarks.

II. RELATED WORK

A. COGNITIVE PERSPECTIVES ON HANDWRITING

Since the advent of writing in human history, literacy education has remained a cornerstone of learning. Writing has served as a primary means for recording knowledge and information, enabling its transmission to future generations and fostering continued societal and intellectual development. Reading and writing are foundational tools of human cognition, and the structural characteristics of individual languages significantly influence how people perceive and process information [13]. Although there is no universally fixed sequence for learning to write, observations of young children indicate that the process typically begins with recognizing the shapes of characters, then moves on to understanding phonetic associations, and eventually progresses to writing characters independently. From a cognitive standpoint, early handwriting integrates visual-motor coordination, orthographic-phonological mapping, and attentional control, which jointly support durable memory traces and transfer to reading and composition.

As computers increasingly permeate daily life, the prevalence of computer-based text input has grown substantially. Despite this shift, the educational value of traditional handwriting—physically writing by hand—remains strongly supported by research. One study demonstrated a positive relationship between handwriting proficiency and academic achievement in both reading and writing [14]. Writing by hand is associated with greater cognitive processing than typing on a keyboard, which has been shown to be associated with improved memory retention [15]. Furthermore, handwriting activities enhance attention, concentration, and cognitive and problem-solving skills [16]. Even in an era defined by the integration of AI into education, the enduring importance of handwriting instruction remains evident across a range of studies.

To improve educational outcomes across subjects, numerous approaches have been developed to enhance handwriting instruction through the use of digital tools. For example, a web-based system that provides automatic error checking for Chinese character writing has been shown to offer more effective learning experiences [17]. Similarly, devices with force feedback mechanisms have been employed to guide learners during pen-based writing activities, yielding positive results [18]. In another study, computerized training programs aimed at improving visual-motor integration were found effective in enhancing the handwriting abilities of students facing difficulties [19]. More recently, tablet-based handwriting applications for kindergarten students have demonstrated learning outcomes comparable to traditional practice in early handwriting education [7]. Importantly, these technology-assisted approaches are interpreted here through a cognitive lens: they seek to scaffold visual-motor integration and attentional control rather than to replace handwriting itself.

B. ONLINE AND OFFLINE CHARACTER RECOGNITION

Optical Character Recognition (OCR) refers to the process of converting scanned or photographed images of handwritten or machine-printed text into machine-readable characters using input devices such as scanners or cameras [1]. Originating from early reading devices for telegraphs and visually impaired individuals in the 1910s, OCR has evolved into a robust field supported by decades of technological advancement and research. Character recognition is commonly categorized as either online or offline, depending on whether time-stamped pen-trajectory data (e.g., $x(t)$, $y(t)$, $p(t)$) are captured (online) or only static images are available (offline).

In the online approach, handwriting is recognized dynamically by analyzing variables such as pen angle, stroke sequence, and movement. This method, which supports low-latency recognition, typically employs pressure-sensitive or electrostatic pens, digitizers that record time-series coordinates, or camera-based systems that can track pen motion and detect writing regions [20]. The recognition process benefits from identifying trajectory points and stroke orders, allowing the system to distinguish among characters. However, online handwriting recognition generally presents greater implementation challenges than its offline counterpart due to its technical complexity. Offline character recognition, by contrast, involves processing static images of handwritten text acquired through scanners, cameras, or screen captures. The digitized characters can then be made searchable, editable, and modifiable. This process typically includes preprocessing the image, segmenting text regions, classifying individual characters, and post-processing the output to produce usable digital text. In this study, as the focus is on recognizing handwritten Korean characters on paper, the offline OCR approach is adopted.

1) AI-BASED CHARACTER RECOGNITION RESEARCH

Artificial intelligence (AI) has significantly impacted the field of Optical Character Recognition (OCR), as it has many other domains. Early efforts in AI-based OCR focused on classifying characters using convolutional neural networks (CNNs), and gradually expanded to incorporate contextual information via recurrent neural networks (RNNs), character region extraction through attention mechanisms, and other advanced techniques [21]. More recently, models based on Vision Transformers (ViTs) have been introduced to enhance both recognition accuracy and processing efficiency.

A critical consideration in OCR research is the distinction between printed and handwritten text. Handwritten characters, unlike printed ones, exhibit considerable variation depending on factors such as time, emotion, context, and individual handwriting style. As a result, handwriting recognition typically yields lower accuracy rates compared to printed text. Furthermore, the accuracy of recognition systems can vary based on document structure: performance differs when dealing with standardized, structured documents

(e.g., official forms) versus unstructured formats such as notebooks or diaries.

In Korean character recognition, Lee et al. achieved accuracy rates of 89.14% and 90.12% using CNN-based models on the PHD08 dataset in real-world conditions [22]. Similarly, Majid and Smith reported a 93.16% accuracy in recognizing Korean consonants and vowels in unstructured document formats by employing region detection and segmentation techniques [12].

Globally, while over 5,000 spoken languages exist, only about 50 writing systems are actively used [23]. A meta-analysis of handwritten character recognition research published between 2000 and 2019 shows a notable increase in activity since 2017, with dominant focus on Roman, Arabic, Indic, and Chinese scripts [24]. In contrast, research on Korean character recognition remains comparatively limited—likely due to its unique use in a single country—contributing to its underrepresentation in global OCR studies.

2) OCR OPEN FRAMEWORK

Advancements in artificial intelligence (AI) have driven substantial progress in character recognition, facilitated by the emergence of open-source frameworks that promote widespread adoption. Prominent examples include Tesseract [1], [25], EasyOCR [26], PaddleOCR [27], and Keras-OCR [28]. Initially developed by Hewlett-Packard in 1985 and open-sourced in 2005, Tesseract has since evolved to leverage long short-term memory (LSTM) neural networks. It supports over 100 languages, including Korean, and is widely adopted due to its high accuracy and ease of integration with Python libraries.

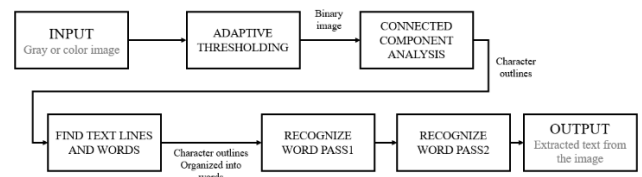


FIGURE 1. Process of tesseract framework [25].

The system first detects character regions using the CRAFT algorithm, and then employs a recognition pipeline that combines ResNet for feature extraction, LSTM for sequence modeling, and a Connectionist Temporal Classification (CTC) decoder. EasyOCR supports over 80 languages and is noted for its rapid processing, high recognition accuracy, and ease of use [26].

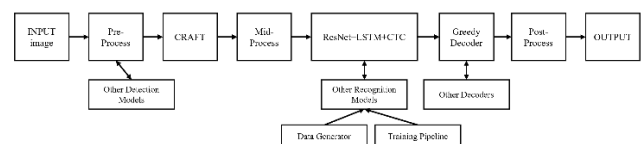


FIGURE 2. Process of EasyOCR framework [26].

Among the open-source OCR frameworks reviewed in this study, EasyOCR demonstrated superior performance in Korean character recognition, particularly in handling handwritten inputs with non-standard spacing and shape variations [11]. In our internal evaluation on classroom worksheet images, while both EasyOCR and PaddleOCR employ similar recognition pipelines—utilizing ResNet for feature extraction, LSTM for sequence modeling, and a CTC decoder—EasyOCR showed greater robustness when tested with actual handwriting samples collected from elementary school students. Its flexible configuration, lighter weight, and ease of integration via Python also contributed to its suitability for educational applications. Accordingly, the present study adopted EasyOCR as the foundation for developing the proposed AI-powered handwriting application.

C. EDUCATIONAL IMPLICATIONS OF AI-BASED LEARNING SUPPORT

There are various perspectives on AI education, including teaching foundational AI and data analysis principles, using AI tools across subjects, and leveraging AI to analyze learners and support personalized learning experiences [9]. Zawacki-Richter et al. classify the educational utilization of AI into four key areas: analysis and prediction, measurement and evaluation, adaptive systems and personalization, and intelligent tutoring systems [10]. These applications enable the prediction of learner characteristics, the validation of assessments, and the design of intelligent systems that provide adaptive and personalized learning. Such technologies help address the limitations of traditional classrooms, where one teacher cannot fully identify and respond to the diverse needs of all students, particularly in the development of foundational skills. In early handwriting, AI support is best positioned as teacher-augmenting formative assessment rather than a replacement for instruction. AI can therefore function as a learner-facing progress signal that augments teacher-led formative assessment; it does not replace teacher judgment. When students receive immediate feedback and clearly understand what steps are needed for improvement, both educational efficiency and motivation can be significantly enhanced. In the present study, feedback is score-based only (overall accuracy per worksheet), with no syllable-block or jamo-level overlays (“character-level highlights”); accordingly, the tool provides a low-latency progress signal that teachers and learners can act upon. Beyond performance, we examine the affective domain (motivation and flow) using established instruments and analyze short open-ended perceptions from students and teachers; responses were thematically coded with double-coding, and inter-rater agreement is reported in Methods.

III. PROPOSED SYSTEM DESIGN

This study developed and implemented a comprehensive educational system leveraging an AI-powered Optical Character Recognition (AI-OCR) engine to support personalized handwriting instruction. The system is delivered as a mobile

application used in class to capture worksheet photos and return prompt, score-based feedback (overall accuracy) within the session; no per-character overlays are shown to learners.

A. OCR SYSTEM INTERACTION FLOW

The system architecture consists of three core components: a mobile client, a RESTful API server, and a dedicated machine learning (ML) server for OCR inference. This modular design ensures scalability, low-latency processing, and interoperability. The complete operational workflow is illustrated in Fig. 3.

The process begins when the mobile client captures an image of the user’s handwritten text. This image is transmitted via HTTP to the API server, which orchestrates the request by forwarding it to the ML server. The ML server then performs the recognition task and returns the analysis to the API server. Finally, the result is relayed back to the mobile application, offering the student a single overall accuracy score (0–100); no syllable-block or jamo-level highlights are displayed.

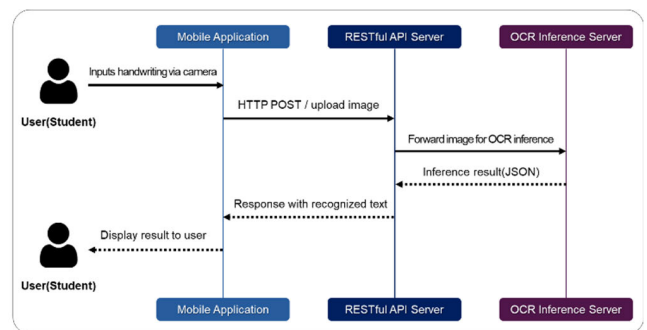


FIGURE 3. OCR system workflow.

1) MODEL ARCHITECTURE AND MODIFICATIONS

The handwriting recognition engine is fundamentally built upon EasyOCR, a high-performance, Python-based open-source framework. The standard EasyOCR pipeline comprises the CRAFT detector (VGG-16 backbone) for text detection and a ResNet-BiLSTM-CTC architecture for sequence recognition. However, this default architecture showed limitations in handling the variable-length structure and the compositional syllable-block structure and shape/spacing variability of handwritten Korean syllables. To address these challenges, the default ResNet-based feature extractor was replaced with a custom lightweight architecture, designed to improve sequence modeling efficiency and reduce computational overhead.

2) DATA PREPARATION AND TRAINING

To ensure robustness and generalizability, the modified model was extensively fine-tuned using a hybrid dataset. This dataset was constructed from two main sources:

- **Synthetic Data:** Korean word and sentence images were synthetically generated using the Text Recognition Data

Generator. These images were rendered in 109 stylistic variants of Naver Nanum fonts (Brush Script and Handwriting) and augmented with skewing, distortion, and blurring to simulate natural variability.

- **Real-world Data:** To incorporate authentic handwriting patterns, the publicly available AI Hub dataset of handwritten Korean characters was also used.

All images were processed through a structured preprocessing pipeline, including grayscale conversion, Gaussian filtering for noise suppression, and Canny edge detection to isolate character boundaries. This hybrid training strategy enhanced the model's ability to generalize across diverse writing styles, including variations in stroke order, spacing, and personal penmanship.

B. EDUCATIONAL APPLICATION UI/UX AND IMPLEMENTATION

To ensure accessibility and low-latency, in-class interaction for young learners, the system was developed and deployed as a mobile application for the Android platform. A modern user interface was built using the Kotlin language and the Jetpack Compose toolkit.

The system is implemented based on a client-server architecture to ensure robust performance and maintainability. This server-side processing allows for the use of a more powerful and complex OCR model than would be feasible for on-device inference, supporting higher accuracy and easier model updates. It also centralizes the machine learning logic, making the model easier to update and manage without requiring users to update the application.

The application's user workflow is designed to be intuitive and pedagogically sound, guiding the learner through a structured practice session. The process unfolds as follows:

- **Mode Selection:** The learner initiates a session by selecting one of two primary modes from the main screen. In 'Dictation Mode,' the application provides a target word for the learner to write. In 'Spelling Check Mode,' the learner undertakes a level-based spelling test. The content for these tests is curated based on developmental stages of orthographic knowledge, following the guidelines provided by the National Institute of the Korean Language.
- **Handwriting Input and Capture:** Instead of direct input on a digital canvas, the system intentionally adopts a traditional writing method. The learner writes the word on physical paper using a real pencil. This pedagogical choice is grounded in the developmental stage of young learners, particularly concerning their fine motor skills. Unlike using a stylus on a slippery glass screen, the tactile feedback and physical resistance from paper and pencil are crucial for developing proper grip, pressure control, and the sensory experience of letter formation. This method also maintains consistency with the authentic classroom environment, maximizing the transfer of learning. Once written, the handwriting image is captured for analysis. Here, the system employs

a teacher-led capture method. Variables that can arise when a young learner captures the photo themselves—such as uneven lighting, inconsistent focus, and camera shake—can significantly impact the accuracy of the OCR analysis. To minimize this data quality uncertainty and ensure the reliability of the analytical results, a trained teacher manages the capture process. This approach guarantees the acquisition of consistent images under optimal conditions, allowing the system to focus its analysis purely on the learner's handwriting, free from environmental noise.

- **Server-Side Analysis and Immediate Feedback:** Upon image capture, the application transmits it to the backend server. The server first performs OCR to recognize the characters and then conducts a spelling check to verify the word's accuracy. The combined analysis results are sent back to the application, which then delivers a single overall accuracy score (0–100) to the learner; no syllable-block or jamo-level overlays ("character-level highlights") are displayed. Teachers may optionally use the score for formative guidance.

This closed-loop feedback mechanism is designed to function as an instructional scaffold, promoting self-directed learning and sustained engagement by transforming handwriting practice into an interactive and responsive experience. The entire pedagogical loop, from task initiation to feedback delivery, is visually detailed in Fig. 4.

IV. EXPERIMENTAL SETUP AND METHODOLOGY

A. PARTICIPANTS

To assess the educational effectiveness of the proposed AI-OCR tool, an experimental study was conducted with 91 Korean first-grade elementary school students (47 boys and 44 girls, ages 7–8) at a public elementary school in Buk-gu, Daegu, South Korea. The study was approved by the school principal. Consistent with the Declaration of Helsinki [29], written informed consent was obtained from parents/guardians, and all personally identifiable information was de-identified prior to analysis. Participation was voluntary, and students could withdraw without penalty.

B. EXPERIMENTAL DESIGN AND PROCEDURE

Participants were organized into five first-grade classes that were randomly assigned at the class level (cluster randomization) to the experimental ($k = 2$, $n = 46$) or control condition ($k = 3$, $n = 45$); students were nested within classes. The study comprised three phases—Pre-test, Intervention, and Post-test—across 10 instructional days (Pre-test on Day 0; Intervention on Days 1–10; Post-test on Day 10 after practice) (see Fig. 5).

In the Pre-test phase (Day 0), both groups completed the EduFlow-2 and ARCS instruments to establish baseline measures.

During the Intervention phase (Days 1–10), all students practiced handwriting with the AI Hangul Writing Teacher application within a 40-min class period split into 30 min



FIGURE 4. Key features and workflow of the application: (a) selecting a learning mode (Dictation Mode or Spelling Check Mode); (b) capturing handwriting on paper (teacher-led); (c) selecting a difficulty level for dictation or spelling; and (d) displaying an overall accuracy score (0–100).

of writing and 10 min of capture/upload. Each day targeted 50 Hangeul syllable blocks (comprising 2–3-syllable words), delivered in two sets (25 + 25) with an optional 1–2-min micro-break. Students wrote on paper worksheets; during the final 10 min, a teacher-operated capture station photographed one image per student and uploaded it to the app.

As illustrated in Fig. 6, the intervention was implemented in the context of a regular handwriting lesson. Students practiced writing Hangeul characters on worksheets, teachers supported the scanning and uploading process using tablet devices, and the classroom environment shows that the activity was embedded within authentic instructional practice (see Fig. 6).

The only difference between groups was the availability of feedback during practice: the experimental group received prompt feedback after submission (overall accuracy 0–100 per worksheet; no per-character overlays), whereas the control group used a measurement-only mode in which accuracy was computed server-side but not displayed during practice. Accuracy was computed as in (1).

$$\text{Accuracy (\%)} = \frac{N(\text{correct})}{N(\text{attempted})} \times 100 \quad (1)$$

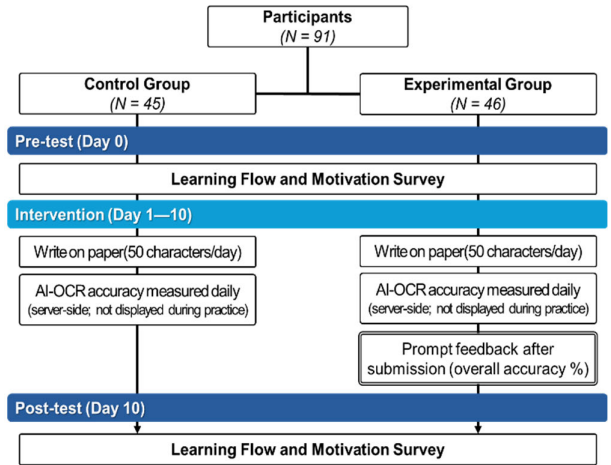


FIGURE 5. Study design and group conditions: feedback vs. measurement-only across a 10-day intervention.

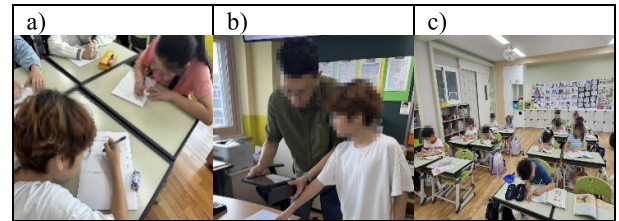


FIGURE 6. Classroom implementation of the AI Hangeul Writing Teacher. (a) Students individually practicing Hangeul characters during the handwriting session. (b) Teacher assisting a student with scanning the handwritten worksheet using the OCR-based application. (c) Classroom context of the handwriting lesson, showing multiple students engaged in the activity.

where $N_{\text{attempted}}$ was defined at the ROI level using ink-density and connected-component thresholds.

Students in the control group viewed their accuracy results only at Post-test. A session was considered valid if students attempted $\geq 80\%$ of items ($\geq 40/50$) within the 30-min writing block; otherwise, outcomes for that day were coded as missing due to the time cap (writing). Upload failures were not treated as missing; same-day re-uploads were permitted. Daily accuracy logs were stored server-side for both groups.

In the Post-test phase (Day 10), both groups again completed the EduFlow-2 and ARCS instruments and had their handwriting accuracy assessed by the AI-OCR system. Additionally, the experimental group completed a brief qualitative survey on user experience and the perceived usefulness of the feedback.

C. MEASUREMENT INSTRUMENTS

Measures. To evaluate the system's impact, we used four instruments: (1) Handwriting Accuracy (primary outcome), (2) Learning Flow (EduFlow-2) [30], (3) Academic Motivation (ARCS-based) [31], and (4) Qualitative feedback.

1) HANDWRITING ACCURACY (PRIMARY OUTCOME)

Daily accuracy was computed server-side by the AI-OCR model from each worksheet (50 items/day). A Hangeul

TABLE 1. Summary of key outcomes (post-test unadjusted means; full ANCOVA in Appendix).

Outcome	Experimental (M)	Control (M)	Δ (Exp–Ctrl)	p
Handwriting accuracy (Day 10, %)	95	73	+22 pp	—
EduFlow-2 (1–7)				
Cognitive Control	5.92	4.51	+1.41	< .001
Immersion / Time Transformation	5.90	4.51	+1.39	< .001
Loss of Self-Consciousness	5.88	4.51	+1.37	< .001
Autotelic Experience	5.99	4.49	+1.50	< .001
ARCS (1–5)				
Attention	4.53	2.83	+1.70	< .001
Confidence	4.51	2.62	+1.89	< .001
Satisfaction	4.57	2.66	+1.91	< .001
Relevance	2.91	2.43	+0.48	< .001

Notes. This table summarizes unadjusted post-test means for readability. Primary analyses used ANCOVA (post ~ group + pre) with class-level adjustment; full ANCOVA adjusted means are reported in Table IV (Appendix A) for EduFlow-2 and in Table V (Appendix A) for ARCS. Scale reliabilities and descriptive statistics are provided in Table III (Appendix A).

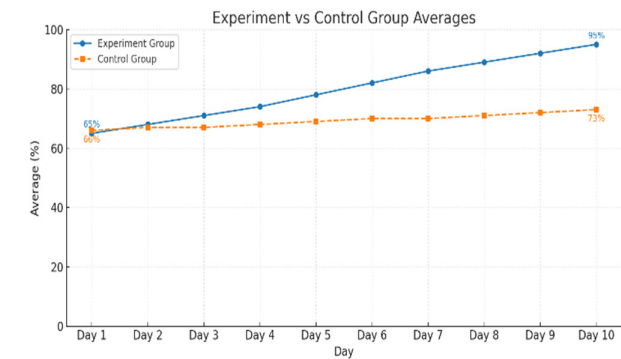


FIGURE 7. Ten-day trends in AI–OCR handwriting accuracy for experimental (feedback) and control (measurement-only) groups.

syllable block was scored correct if the recognized label exactly matched the target. We defined “attempted” at the region of interest (ROI) level using ink-density and connected-component thresholds. The daily accuracy score (0–100) was calculated as in (1), representing the percentage of correctly recognized syllable blocks out of the 50 items on each worksheet. Sessions were considered valid if $\geq 40/50$ items were attempted within the 30-min writing block (see Sec. III-B).

2) LEARNING FLOW (PRE/POST)

A child-adapted version of EduFlow-2 was administered [30]. The scale covers four dimensions (cognitive control; immersion/time transformation; loss of self-consciousness; autotelic experience). Wording was simplified for first-grade readability and administered on a child-adapted 7-point pictorial Likert scale (1–7; higher scores indicate greater flow) with smiley anchors; a brief practice was used to ensure comprehension. Cronbach’s α is reported at pre- and post-test.

3) ACADEMIC MOTIVATION (PRE/POST)

Motivation was measured with a child-adapted, ARCS-based scale assessing Attention, Relevance, Confidence, and Satisfaction [32], administered on a child-adapted 5-point pictorial

TABLE 2. Descriptive statistics of student sentiment (Experimental group).

Method	Score	Mean	SD	Min	Max
Hu and Liu VADER	Polarity (−100...100)	11.39	9.23	−22.22	33.33
	Positive	0.19	0.18	0.00	0.70
	Negative	0.01	0.05	0.00	0.47
	Neutral	0.79	0.19	0.29	1.00
Multilingual Sentiment SentiArt	Compound (−1...1)	0.27	0.28	−0.66	0.86
	Polarity (−100...100)	19.53	15.43	−40.00	60.00
	Sentiment (Valence)	0.26	0.29	−0.76	1.25
	Anger	0.56	0.32	−0.21	1.43
	Fear	0.42	0.34	−0.37	1.21
	Disgust	0.27	0.30	−0.46	0.91
	Happiness	0.94	0.39	0.03	2.26
	Sadness	0.73	0.35	−0.27	1.80
	Surprise	0.28	0.38	−0.48	1.17

Notes. Hu and Liu Polarity, Multilingual Sentiment Polarity = −100...100; VADER Positive/Negative/Neutral are proportions (0...1); VADER Compound = −1...1; SentiArt scores are continuous (typical range $\approx -2...+2$). Values are computed on translated (KR→EN) responses, $n = 46$. Due to rounding, VADER proportions may not sum exactly to 1.00.

Likert scale (1–5; higher scores indicate greater motivation) with smiley anchors; a brief practice was used to ensure comprehension. Cronbach’s α is reported at pre- and post-test. Scale reliabilities and descriptive statistics (pre/post) for both EduFlow-2 and ARCS are summarized in Table 3 in Appendix.

4) QUALITATIVE FEEDBACK (POST)

A short open-ended survey was administered to students in the experimental group (user experience; perceived usefulness of the feedback) and to all five classroom teachers (instructional relevance, usability, and classroom integration). Responses were thematically coded by two raters; inter-rater agreement (Cohen’s κ) is reported [33].

V. RESULTS

A. QUANTITATIVE EVALUATION

1) IMPROVEMENT IN HANDWRITING RECOGNITION ACCURACY

As shown in Fig. 7, the experimental group exhibited a steady increase in AI–OCR handwriting accuracy over the

TABLE 3. Reliabilities and descriptive statistics (PRE/POST) for EduFlow-2 and ARCS.

Constructs	Sub-components	survey	Group	Cronbach's alpha	MEAN	SD	t	p
Flow	Cognitive control	pre	Experimental	0.71	4.01	0.61	-0.43	0.67
			Control		4.07	0.67		
		post	Experimental	0.80	5.92	0.52	15.81	< .001
			Control		4.51	0.30		
	Immersion and Time transformation	pre	Experimental	0.73	4.03	0.62	0.27	0.792
			Control		3.99	0.70		
		post	Experimental	0.80	5.90	0.52	14.98	< .001
			Control		4.51	0.34		
	Loss of self-consciousness	pre	Experimental	0.77	4.01	0.69	0.00	1.00
			Control		4.01	0.70		
		post	Experimental	0.77	5.88	0.46	16.86	< .001
			Control		4.51	0.30		
Motivation	Autotelic experience	pre	Experimental	0.87	3.96	0.78	-0.71	0.48
			Control		4.07	0.69		
		post	Experimental	0.83	5.99	0.57	15.48	< .001
			Control		4.49	0.31		
	Attention	pre	Experimental	0.75	2.69	0.40	-0.94	0.35
			Control		2.80	0.73		
		post	Experimental	0.94	4.53	0.46	14.32	< .001
			Control		2.83	0.65		
	Relevance	pre	Experimental	0.65	2.58	0.29	-0.83	0.40
			Control		3.07	0.61		
		post	Experimental	0.83	2.91	0.87	3.51	< .001
			Control		2.43	0.31		
	Confidence	pre	Experimental	0.64	2.92	0.36	0.96	0.34
			Control		2.81	0.65		
		post	Experimental	0.93	4.51	0.29	26.23	< .001
			Control		2.62	0.39		
	Satisfaction	pre	Experimental	0.62	2.92	0.36	-1.39	0.17
		post	Experimental	0.93	4.57	0.32	23.72	< .001

10-day period, rising from a mean of 65% (Day 1) to 95% (Day 10). By contrast, the control group—for whom accuracy was computed server-side but not displayed during practice (measurement-only)—showed a modest gain from 66% to 73%. Baseline levels were comparable, and the rate of improvement was larger in the experimental group ($\approx +30$ pp) than in the control group ($\approx +7$ pp), a pattern computed over valid sessions ($\geq 40/50$ attempted; see Sec. IV) and consistent with prompt, score-based post-submission feedback (overall accuracy only) on handwriting performance.

2) FLOW EXPERIENCE

Flow was measured with a child-adapted version of EduFlow-2 at pre- and post-test (1–7; higher scores indicate greater flow) [30]. No between-group differences were observed at pre-test. At post-test, the experimental group scored higher than the control group across the four dimensions (unadjusted post-test two-sided tests, $p < .001$):

Cognitive Control ($M = 5.92$ vs. 4.51 ; $t = 15.81$), Immersion/Time Transformation (5.90 vs. 4.51 ; $t = 14.98$), Loss of Self-Consciousness (5.88 vs. 4.51 ; $t = 16.86$), Autotelic Experience (5.99 vs. 4.49 ; $t = 15.48$). Primary analyses using ANCOVA (post $\sim$ group + pre) with class-level adjustment corroborated these effects; Adjusted means and effect sizes (ANCOVA with pre-test as covariate) are summarized in Table 1; full ANCOVA outputs are provided in Table 4 in Appendix.

3) MOTIVATIONAL OUTCOMES (ARCS)

Academic motivation (ARCS; 1–5 scale) [32] also favored the experimental group at post-test: Attention ($M = 4.53$ vs. 2.83 ; $t = 14.32$, $p < .001$), Confidence (4.51 vs. 2.62 ; $t = 26.23$, $p < .001$), Satisfaction (4.57 vs. 2.66 ; $t = 23.72$, $p < .001$), Relevance (2.91 vs. 2.43 ; $t = 3.51$, $p < .001$). No gender effects were detected. By academic achievement, a difference emerged for Confidence ($F = 4.34$, $p < .05$), with higher-performing students showing

TABLE 4. EduFlow-2: post-test adjusted means (ANCOVA with pre-test as covariate) by group.

Constructs	Sub-components	Gender	N	MEAN	SD	t	p
Flow	Cognitive control	Man	22	5.99	0.57	0.83	0.41
		Woman	23	5.86	0.48		
	Immersion and Time transformation	Man	22	5.79	0.42	-1.38	0.18
		Woman	23	6.00	0.60		
	Loss of self-consciousness	Man	22	5.92	0.38	0.61	0.54
		Woman	23	5.84	0.52		
Motivation	Autotelic experience	Man	22	6.00	0.44	0.17	0.87
		Woman	23	5.97	0.69		
	Attention	Man	22	4.47	0.49	-0.80	0.43
		Woman	23	4.58	0.43		
	Relevance	Man	22	2.89	0.84	-0.13	0.90
		Woman	23	2.93	0.91		
	Confidence	Man	22	4.53	0.30	-0.43	0.67
		Woman	23	4.49	0.28		
	Satisfaction	Man	22	4.55	0.33	-0.50	0.62
		Woman	23	4.59	0.32		

larger gains. Results were consistent in ANCOVA models adjusting for pre-test with class-level correction; adjusted means and effect sizes are summarized in Table 1, with full ANCOVA outputs in Table 5 in Appendix A.

B. QUALITATIVE EVALUATION

To complement the quantitative findings, this study collected qualitative data from students and teachers through open-ended surveys and sentiment analysis. The analysis focused on the perceived usability, effectiveness, and experiential value of the AI-OCR tool to provide a deeper understanding of its impact on learner engagement and its pedagogical implications.

1) STUDENT PERCEPTIONS AND SENTIMENT ANALYSIS (EXPERIMENTAL GROUP)

Post-test, students in the experimental group provided short open-ended comments about their experience with the tool (e.g., “fun,” “interesting,” “I tried to raise my score”). Responses were written in Korean, translated into English, and analyzed in Orange3 using four lexicon-based models: Hu and Liu [17], VADER [34], Multilingual Sentiment [8], and SentiArt [35]. On Orange3’s Multilingual Sentiment scale (−100...100), student comments showed a mild positive polarity ($M = 19.53$, $SD = 15.43$; range = −40.00...60.00; $n = 46$), consistent with VADER compound ($M = 0.27$). Descriptive statistics for all models are summarized in Table 2. Given known limitations of lexicon methods with early-grade language and translation, these results are interpreted as supportive but descriptive evidence of engagement.

2) TEACHER FEEDBACK

Post-intervention, all five participating classroom teachers provided short open-ended feedback via an online survey.

Teachers generally endorsed the tool’s value for self-directed practice and engagement via prompt feedback. Concerns centered on over-reliance on technology, the need for alignment with curriculum standards, and requests for more granular feedback beyond a single overall accuracy percentage (outside the current design scope) as well as improved capture guidance/OCR robustness under suboptimal lighting.

VI. CONCLUSION

This work developed and classroom-deployed the AI Hangul Writing Teacher, an AI-OCR-based tool designed to deliver scalable, learner-specific, score-based feedback (overall accuracy, 0–100) in early handwriting instruction. The system integrates a customized EasyOCR model with a teacher-operated capture workflow (paper worksheets → server-side recognition) and provides prompt, post-submission feedback to students showing a single overall accuracy score; no per-character (syllable/jamo) overlays are displayed.

In a 10-day, class-randomized classroom study with $N = 91$ first-grade students, the experimental (feedback) group showed a steady improvement in AI-OCR handwriting accuracy from 65% to 95%, whereas the control (measurement-only) group increased from 66% to 73%. Post-test flow and motivation scores also favored the experimental group, and student comments indicated predominantly positive sentiment. Together, these results are consistent with the value of prompt, AI-supported, score-based feedback for strengthening early handwriting performance and engagement while maintaining a paper-based classroom workflow.

This study suggests that thoughtfully designed AI-OCR systems can function as effective, scalable supports for foundational literacy instruction. Key limitations include the single-school setting, short study duration (10 days), and the use of an overall-accuracy-only feedback design (no per-character highlights). Because OCR accuracy reflects

TABLE 5. ARCS: Post-test adjusted means (ANCOVA with pre-test as covariate) by group.

Constructs	Sub-components	Academic Level	N	MEAN	SD	F	p	Scheffé
Flow	Cognitive control	High(a)	6	6.17	0.28	0.77	0.47	-
		Medium(b)	28	5.88	0.48			
		Low(c)	11	5.88	0.70			
	Immersion and Time transformation	High(a)	6	6.00	0.42	1.88	0.16	-
		Medium(b)	28	5.98	0.43			
		Low(c)	11	5.64	0.71			
	Loss of self-consciousness	High(a)	6	6.06	0.44	0.51	0.60	-
		Medium(b)	28	5.85	0.37			
		Low(c)	11	5.88	0.65			
	Autotelic experience	High(a)	6	5.83	0.55	0.24	0.79	-
		Medium(b)	28	6.01	0.53			
Motivation	Attention	High(a)	6	4.78	0.40	1.65	0.20	-
		Medium(b)	28	4.54	0.46			
		Low(c)	11	4.36	0.46			
	Relevance	High(a)	6	3.28	1.02	0.61	0.55	-
		Medium(b)	28	2.86	0.88			
		Low(c)	11	2.85	0.78			
	Confidence	High(a)	6	4.72	0.25	4.34	0.02	a<c
		Medium(b)	28	4.54	0.29			
		Low(c)	11	4.33	0.21			
	Satisfaction	High(a)	6	4.78	0.27	1.83	0.17	-
		Medium(b)	28	4.51	0.33			
		Low(c)	11	4.61	0.29			

visual-form correctness but does not capture stroke order, pen pressure, spacing consistency, or motor fluency, the present findings should be interpreted as improvements in recognized-form accuracy rather than holistic handwriting quality. Future work will extend evaluation to diverse schools and grade levels, investigate more granular formative feedback—including jamo-level or stroke-pattern highlighting, vowel-slant detection, or batchim-omission cues—examine long-term retention, and optimize the capture/processing pipeline and teacher dashboards. Assessing generalization across handwriting styles and corroborating OCR accuracy with human double-ratings at larger scale will further strengthen external validity.

APPENDIX

See Tables 3–5.

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